

3.1W小型便捷低功耗I²S输入单声道D类音频功率放大器

3.1W Tiny, Plug-and-Play, Low Consumption, I²S Input, Mono Class D Audio Amplifier

■ FEATURES

- Plug-and-Play with a single bypass capacitor
- Power Supply VDD: 2.5-5.5V;
- Low Quiescent Current: 2.5mA @VDD = 3.6V
- Low Output Noise: 15uV
- Low Pop Noise
- Flexible Audio I/O
 - I²S (Left, Right, or Left/2+Right/2 Output), 16bit, 24bit or 32 bit; TDM input, 16bit or 32bit
 - 8k, 16k, 32k, 44.1k, 48k, 88.2k, 96kHz Sample Rates
 - No MCLK Required
- Output Power
 - 1.5W (V_{DD}=3.6V, R_L=4Ω, THD+N=10%)
 - 3.1W (V_{DD}=5.0V, R_L=4Ω, THD+N=10%)
 - 200mW (V_{DD}=3.6V, R_L=32Ω, THD+N=1%)
- THD+N= 0.02% (P_o=1W, R_L=4Ω, V_{DD} = 3.7V)
- Adjustable Gain: -0.7, 5.3, 8.3, 11.3, 14.3dBV
- Clock Error, Over Current, Undervoltage, and Overtemperature Protection
- Pb-free Packages, WLCSP9, ESOP8, QFN3×3-16L
- 上电即用, 外围极简 (最少仅需一个电源滤波电容)
- 电源供电: VDD 2.5-5.5V;
- 低静态电流: 2.5mA @ VDD = 3.6V
- 低输出噪声: 15uV
- 低pop声
- 灵活的音频输入:
 - I²S (输出左、右或左/2+右/2可选择), 16bit, 24bit 或 32 bit; TDM 输入, 16bit或32bit
 - 8k, 16k, 32k, 44.1k, 48k, 88.2k, 96kHz 采样频率
 - 无需MCLK
- 输出功率:
 - 1.5W (V_{DD}=3.6V, R_L=4Ω, THD+N=10%)
 - 3.1W (V_{DD}=5.0V, R_L=4Ω, THD+N=10%)
 - 200mW (V_{DD}=3.6V, R_L=32Ω, THD+N=1%)
- THD+N= 0.02% (P_o=1W, R_L=4Ω, V_{DD} = 3.7V)
- 增益可调: -0.7, 5.3, 8.3, 11.3, 14.3dBV
- 保护: 时钟错误、过流、欠压、过温保护等
- WLCSP9, ESOP8, QFN3×3-16L封装

■ APPLICATIONS

- OWS and other earphone devices
- Smart Home
- IoT Devices
- Smart Speakers
- Smart Toys
- Gaming Devices
- Other Li-ion cell/5V Devices
- OWS耳机或其他耳机设备
- 智能家居
- 游戏设备
- 智能玩具
- 智能音箱
- IoT设备
- 其他锂电/5V设备

■ DESCRIPTION

The HT517 is an easy-to-use, low-cost, I²S input mono Class D audio amplifier that drives up to continuous 3.1W into 4ohm speaker with 5V power supply, and 180mW into 32ohm speaker with 3.7V power supply.

HT517 integrates an DA converter, the I²S interface supports up to 32bit width, 96kHz sample rate.

The built-in Class D amplifier also integrates spread spectrum function. The gain of the amplifier is also adjustable.

The Device is available in WLCSP9 and ESOP8 package.

HT517是一款低成本的I²S输入单声道输出D类音频功率放大器。其在5V供电时可提供最大3.1W的输出功率; 在3.7V供电时可提供最大180mW的输出功率。

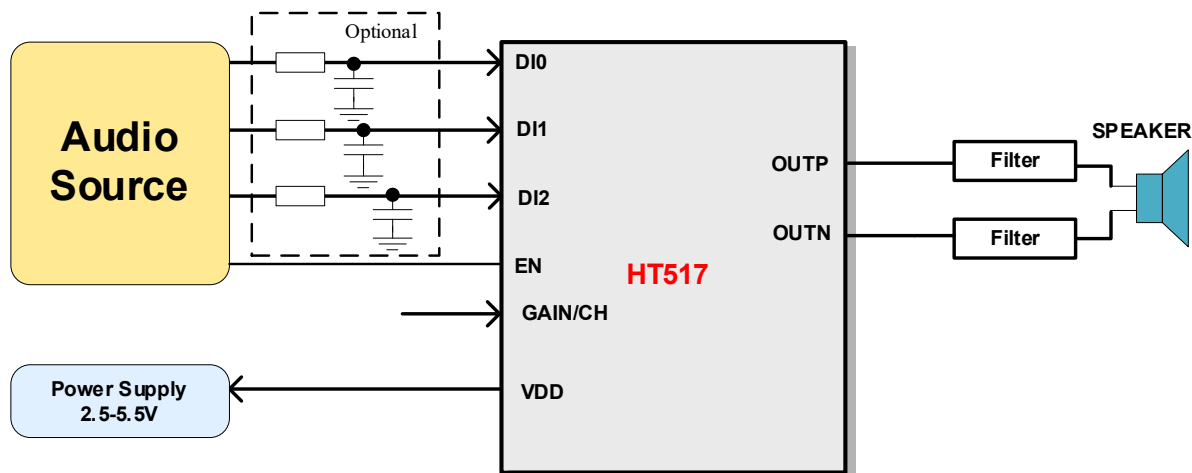
HT517内部集成了DA转换器, 其I²S输入最大支持32-bit字节, 并且可自动监测采样频率, 最高支持96kHz。

HT517集成的D类音频功放还具有增益可调、扩频等功能。

该产品提供WLCSP9和ESOP8封装。

■ TYPICAL APPLICATION

单声道应用 Mono Operation



1. DIx模式设置，通道选择

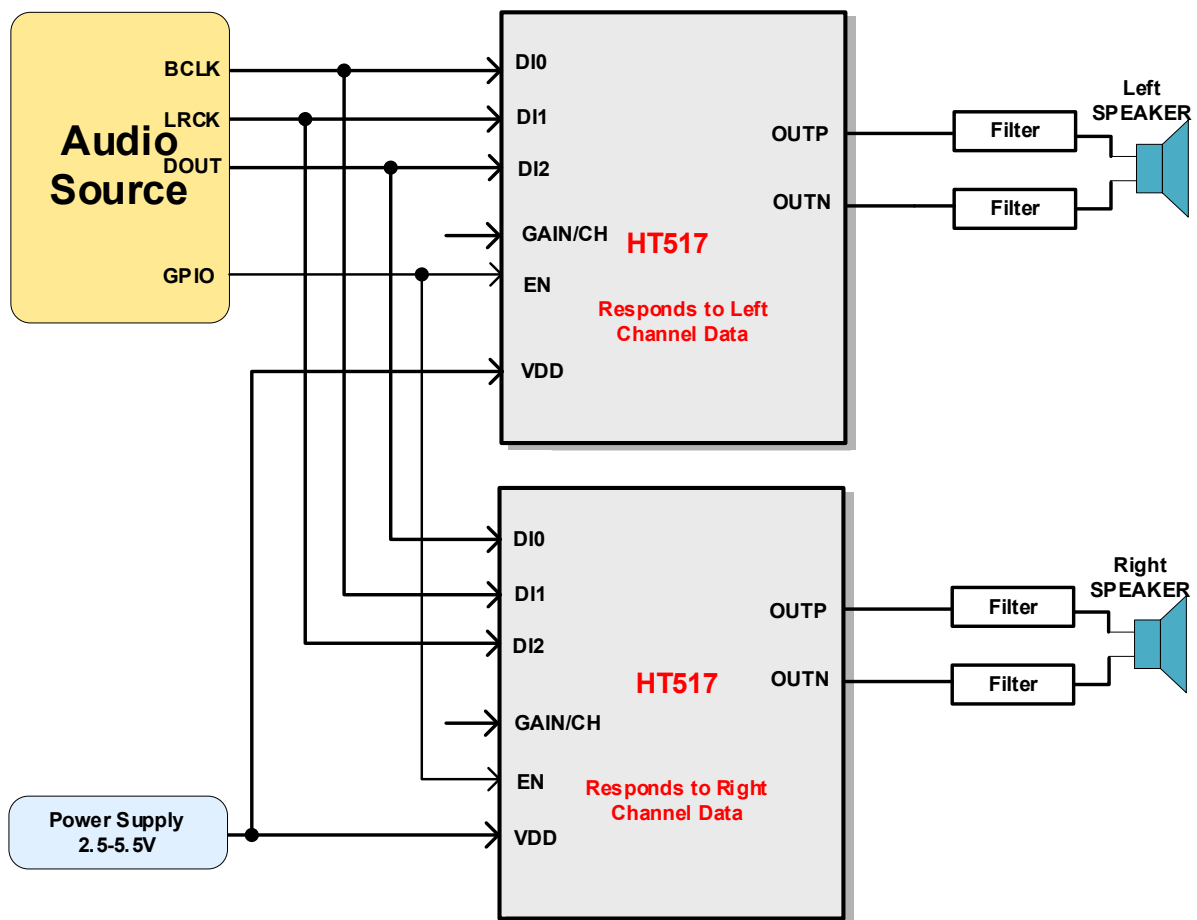
BCLK	LRCK	SDIN	DIx Mode	Channel Selection
DI0	DI1	DI2	A	Left
DI1	DI2	DI0	B	Right
DI2	DI0	DI1	C	Left/2+Right/2

DI0, DI1, DI2可增加RC电路（optional，典型值33R, 10pF），防止走线较长、寄生电感大、波动大时引起的信号异常。以下立体声、TDM等应用时相同。

2. GAIN/CH脚设置，增益选择

GAIN/CH pin	System Gain
Directly connected to GND	14.3dBV
Floating	11.3dBV
Directly connected to VDD	8.3dBV
Connected to VDD thru 100k resistor	5.3dBV
Connected to GND thru 100k resistor	-0.7dBV

立体声应用 Stereo Operation

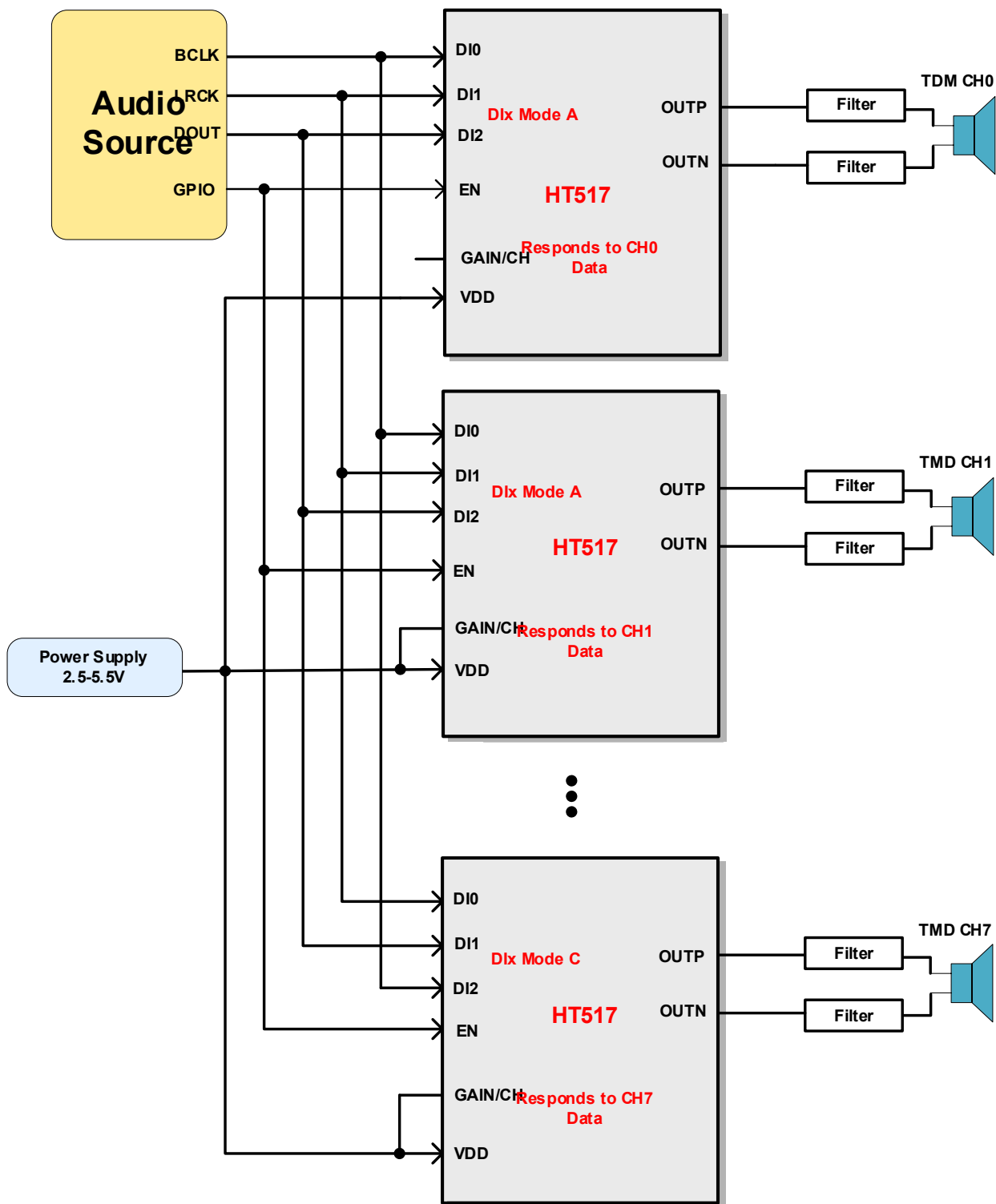


3. GAIN/CH脚设置，增益选择

GAIN/CH pin	System Gain
Directly connected to GND	14.3dBV
Floating	11.3dBV
Directly connected to VDD	8.3dBV
Connected to VDD thru 100k resistor	5.3dBV
Connected to GND thru 100k resistor	-0.7dBV

TDM应用（见下页图），增益固定在14.3dBV TDM Operation, Gain is fixed at 14.3dBV

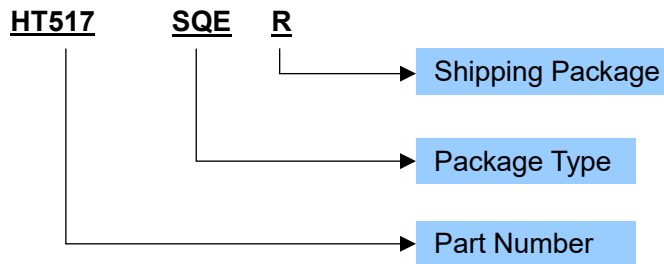
DIx Mode	GAIN/CH	Selected Channel
A	GND	0
A	VDD	1
A	Floating	2
B	VDD	3
B	GND	4
C	GND	5
C	Floating	6
C	VDD	7



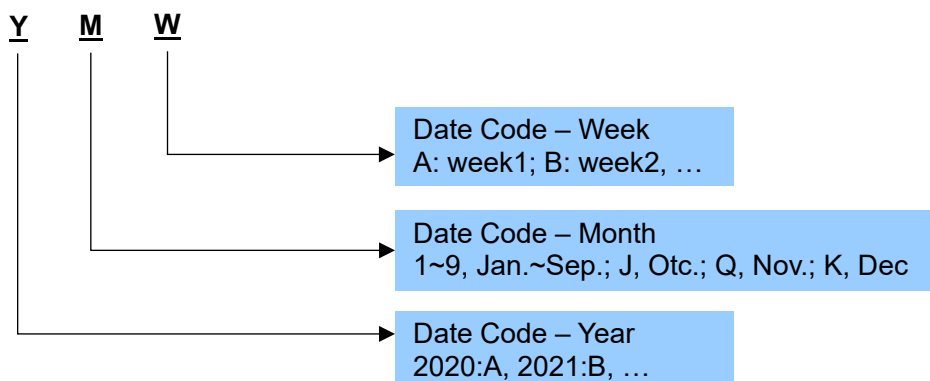
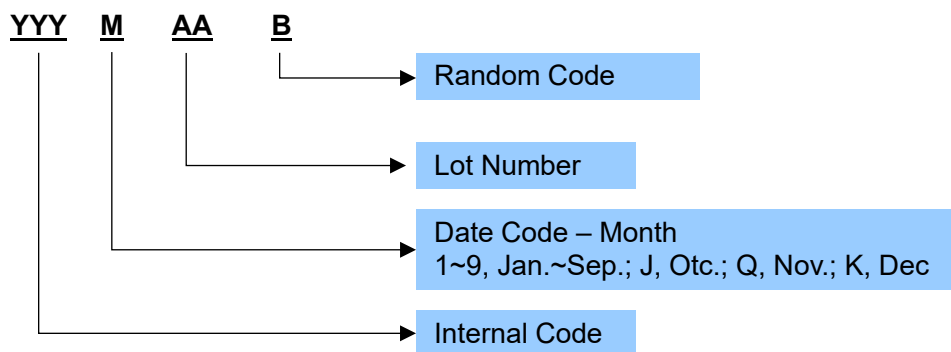
■ ORDERING INFORMATION

Part Number	Package Type	Marking	Operating Temperature Range	MOQ/Shipping Package
HT517WCSR	WLCSP9	517 YMW ¹	-40℃~85℃	5000 / Tape and Reel
HT517SQER	QFN3×3-16L	HT517 YYYMAAB	-40℃~85℃	5000 / Tape and Reel
HT517SPER	ESOP8	HT517 YYYMAAB ²	-40℃~85℃	2500 / Tape and Reel

Ordering Number



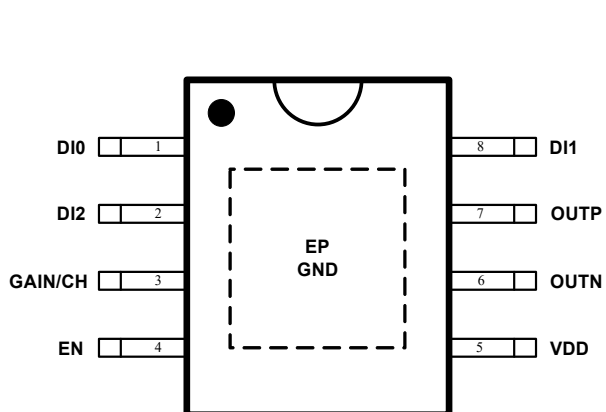
Production Tracking Code



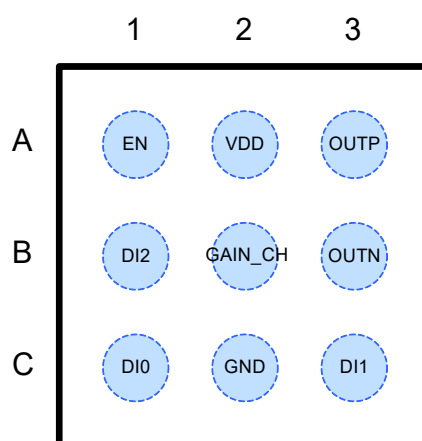
¹ YMW is production tracking code

² YYYMAAB is production tracking code

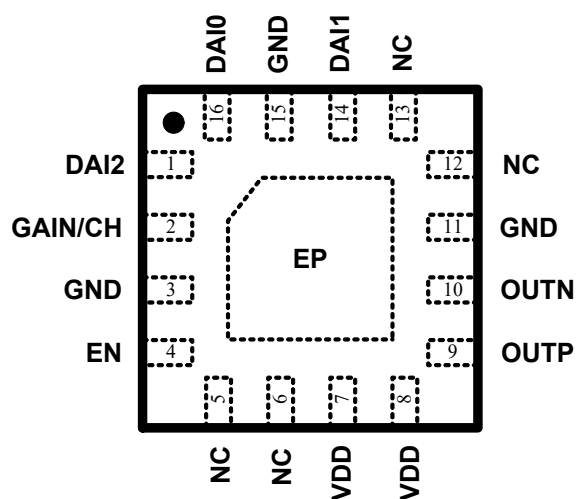
■ TERMINAL CONFIGURATION



ESOP8 Top View



WLCSP9 Top View



QFN3x3-16L Top View

■ TERMINAL FUNCTION

Terminal No.		Name	I/O ¹	Description
ESOP (WLCSP)	QFN			
1 (C1)	16	DI0	I	Digital Audio Interface 0. 数字音频接口 0
2 (B1)	1	DI2	I	Digital Audio Interface 2. 数字音频接口 2
3 (B2)	2	GAIN/CH	I	Gain and channel selection. Select the amplifier gain. Select the channel and DI configuration in TDM mode (in TDM mode, the gain is fixed at 14.3dBV) 增益和通道选择。通过该引脚, 可选择功放的增益, 可选择 TDM 的通道和 DI 引脚的设置
4 (A1)	4	EN	I	Enable pin, when EN pulls low, the device is in Shutdown mode. 使能脚, 接地时器件关闭。
5 (A2)	7,8	VDD	P	Power Supply. 电源
6 (B3)	10	OUTN	O	Negative pin for differential speaker amplifier. 功放输出负端
7 (A3)	9	OUTP	O	Positive pin for differential speaker amplifier. 功放输出正端
8 (C3)	14	DI1	O	Digital Audio Interface 1. 数字音频接口 1
EP (C2)	3, 11	GND	G	Ground pad, also provides thermal connection from the device to the board. 地, 同时为器件散热通道。
--	5,6,12,13, EP	NC	--	No Connection, Connect to GND. 内部无连接, 可连接至地。

¹ I: Input; O: Output; G: Ground; P: Power

■ SPECIFICATIONS¹

● Absolute Maximum Ratings²

PARAMETER	Symbol	MIN	TYP	MAX	UNIT
Power supply voltage range	VDD	-0.3		6	V
Digital Input Voltages (DI0, DI1, DI2)	V _I	-0.3		6	V
All other pins	V _I	-0.3		VDD+0.3	V
Moisture Sensitivity Level (MSL)			MSL3		
Ambient Operating Temperature	T _A	-40		85	°C
Junction Temperature	T _J	-40		150	°C
Storage Temperature	T _{STG}	-40		150	°C

● Recommended Operating Conditions

PARAMETER	Symbol	CONDITION	MIN	TYP	MAX	UNIT
Power supply voltage range	VDD		2.5		5.5	V
Ambient Operating Temperature	T _a		-40	25	85	°C
Minimum Speaker Load	R _L	PVDD<5.5V	3			Ω
		PVDD<4.5V	2			Ω

● I/O pins

PARAMETER	Symbol	CONDITION	MIN	TYP	MAX	UNIT
Input High Voltage	V _{IH1}	DI0, DI1, DI2	0.84			V
Input Low Voltage	V _{IL1}				0.54	V
Input Leakage Current	I _{IH1} , I _{IL1}		-1		+4	μA
Input High Voltage	V _{IH2}	EN	1.0			V
Input Low Voltage	V _{IL2}				0.24	V
Input Leakage Current	I _{IH2} , I _{IL2}		-1		+1	μA

● System

T_A = 25°, VDD = 3.7V, GAIN = 14.3dBV, f_s = F_{LRCK} = 48kHz, f_{BCLK} = 3.72MHz, unless otherwise noted

PARAMETER	Symbol	CONDITION	MIN	TYP	MAX	UNIT
Quiescent current	I _{VDD}	VDD = 5V		2.8		mA
		VDD = 3.7V		2.5		mA
		VDD = 3.3V		2.45		mA
Standby current	I _{Standby}	EN = 1.8V, DIx = 0V		1.5		μA
		EN = 1.8V, DIx no toggling		3.5		μA
Shutdown current	I _{SD}	EN = 0V		0.01		μA
Turn-on time	t _{ON}			1		ms

● Protection Circuitry

PARAMETER	Symbol	CONDITION	MIN	TYP	MAX	UNIT
VDD Undervoltage Error Threshold	U _{VERTH}	VDD Falling		1.9		V
VDD Undervoltage Error Threshold	U _{VEFTH}	VDD Rising		2.0		V
Overtemperature Error Threshold	O _{TE} TH			150		°C
Overtemperature Error Hysteresis	O _{TE} HYS			15		°C
Overtemperature Error Recovery Time	t _{OPT}			100		us
Overcurrent Error Threshold	O _{CE} TH			3.6		A
Overcurrent Error Recovery Time	t _{OCP}			100		us

¹ Depending on parts and PCB layout, characteristics may be changed.

² Stresses beyond those listed under absolute maximum ratings may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under recommended operating conditions is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

● Speaker Amplifier

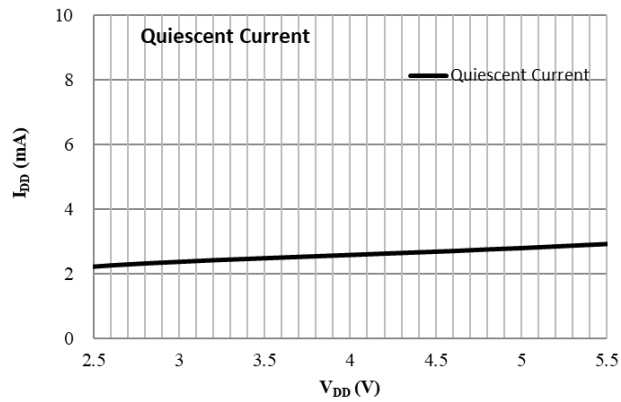
TA = 25°, VDD = 3.7V, GAIN = 14.3dBV, fs = f_{LRCK} = 48kHz, f_{BCLK} = 3.072MHz, unless otherwise noted

PARAMETER	Symbol	CONDITION	MIN	TYP	MAX	UNIT
Idle Channel Noise	V _N	GAIN = 14.3dBV, 20-20kHz, A-Weighted		15		μV
Signal to Noise Ratio (Referenced to THD+N=1%)	SNR	R _{SPK} = 4Ω, Gain = 14.3dBV, 20-20kHz, A-Weighted		107		dB
Speaker Amplifier Gain, I ² S	A _V	I ² S, GAIN/CH = GND; or TDM		14.3		dBV
		GAIN/CH Floating		11.3		dBV
		GAIN/CH = VDD		8.3		dBV
		GAIN/CH = VDD through 100kohm		5.3		dBV
		GAIN/CH = GND through 100kohm		-0.7		dBV
Speaker Amplifier DC Offset	V _{OS}			0.3		mV
Speaker Amplifier Switching Frequency	f _{SPK_AMP}			300		kHz
Spread Spectrum Bandwidth				± 14		kHz
Maximum Instantaneous Output Power Per. Ch.	P _O	VDD = 5 V, R _{SPK} = 4 Ω, THD+N = 1%		2.5		W
		VDD = 5 V, R _{SPK} = 8 Ω, THD+N = 1%		1.4		W
		VDD = 3.7 V, R _{SPK} = 32 Ω, THD+N = 1%		200		mW
		VDD = 5V, R _{SPK} = 4 Ω, THD+N = 10%		3.1		W
		VDD = 5V, R _{SPK} = 8 Ω, THD+N = 10%		1.7		W
Total Harmonic Distortion and Noise	THD+N	VDD = 5 V, R _{SPK} = 4 Ω, P _O = 1 W		0.02		%
Efficiency	η	VDD = 5 V, R _{SPK} = 8 Ω+33uH, THD+N = 10%		90		%

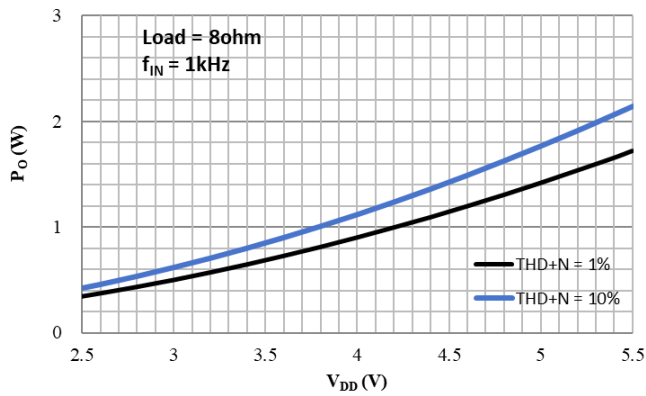
■ TYPICAL OPERATING CHARACTERISTICS

VDD = 5V, GAIN = 14.3dBV, fs = fLRCK = 48kHz, fBCLK = 3.072MHz, unless otherwise specified

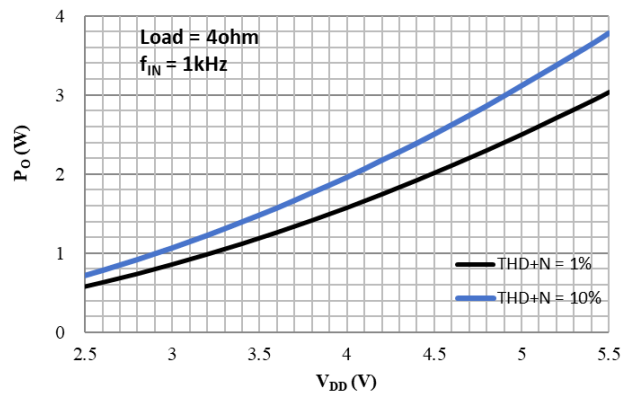
V_{DD} vs I_{DD}



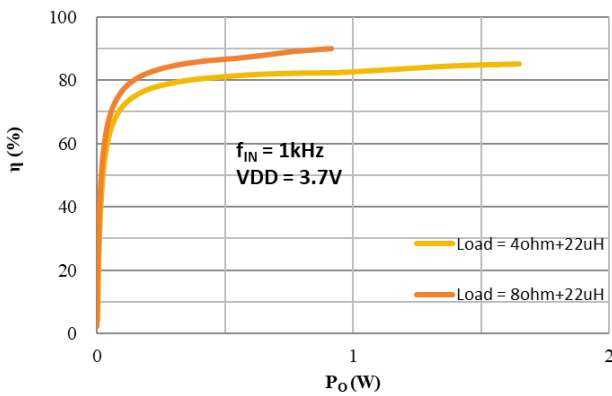
V_{DD} vs P_O



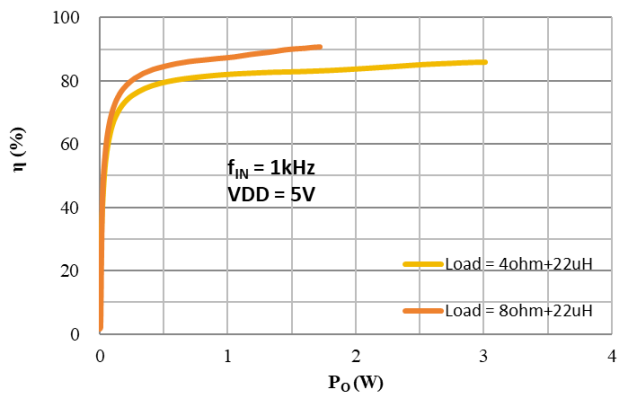
V_{DD} vs P_O

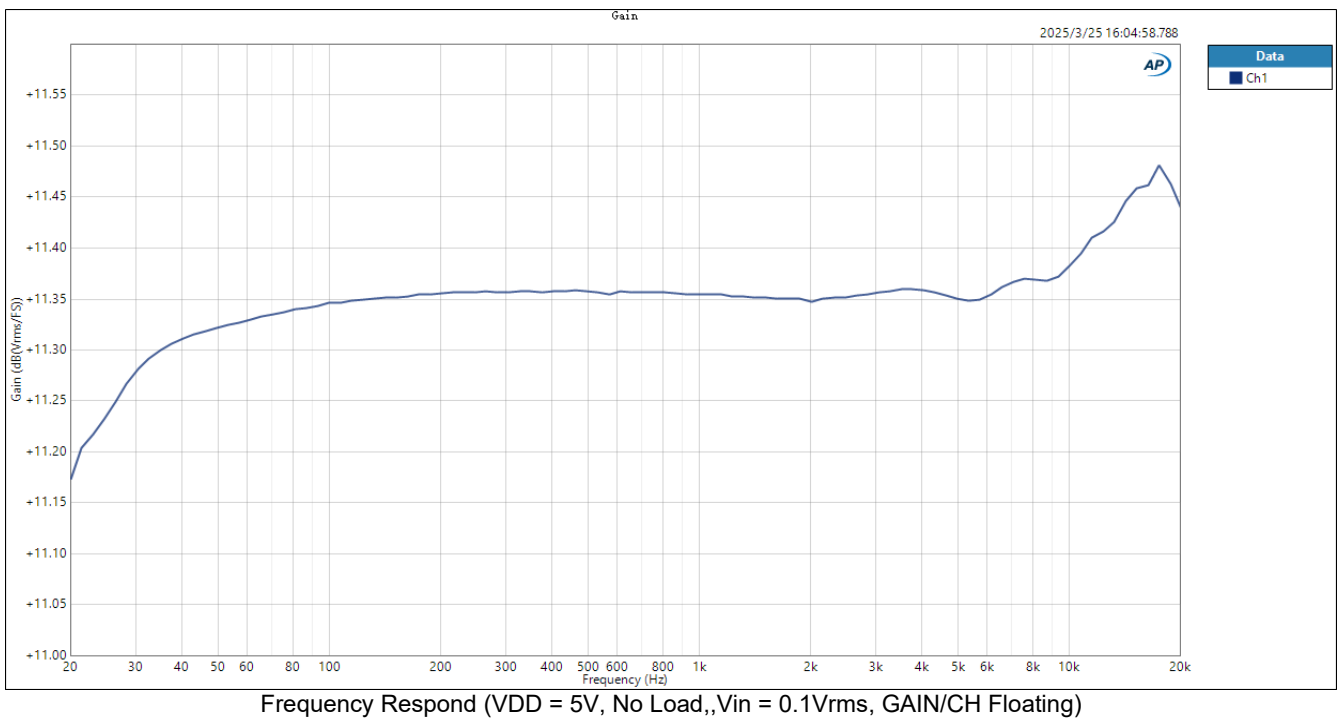
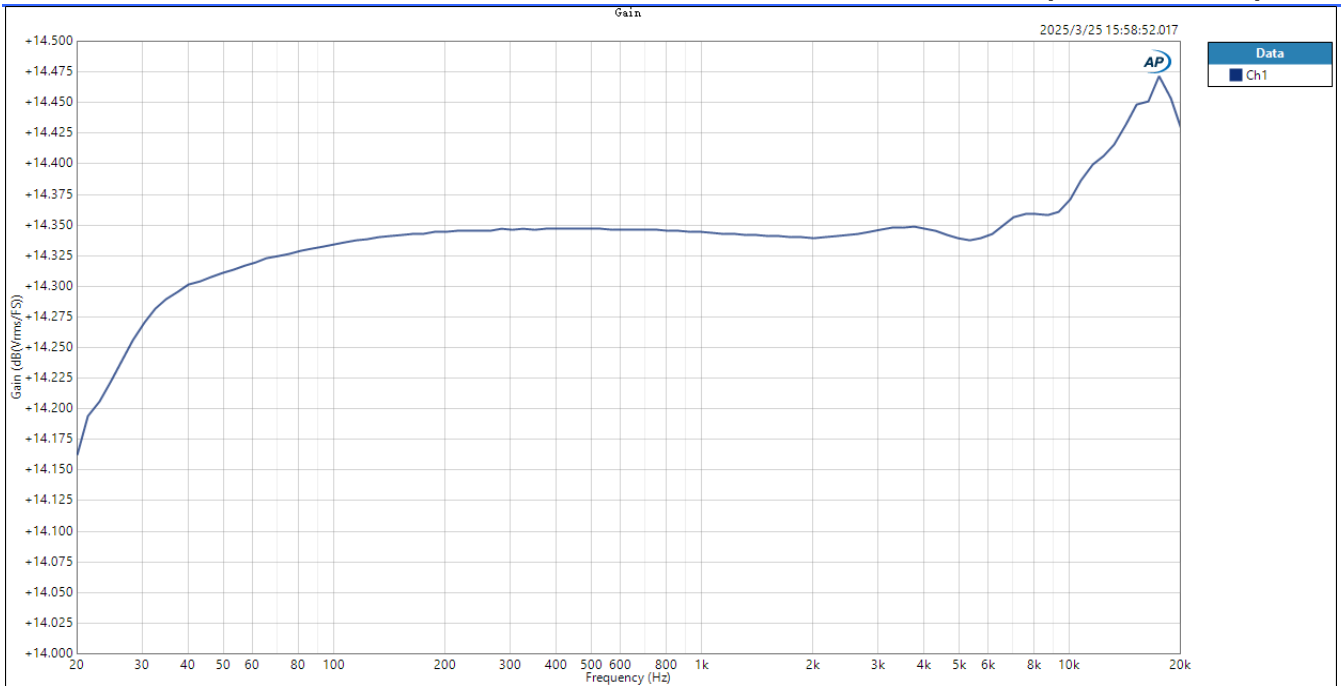


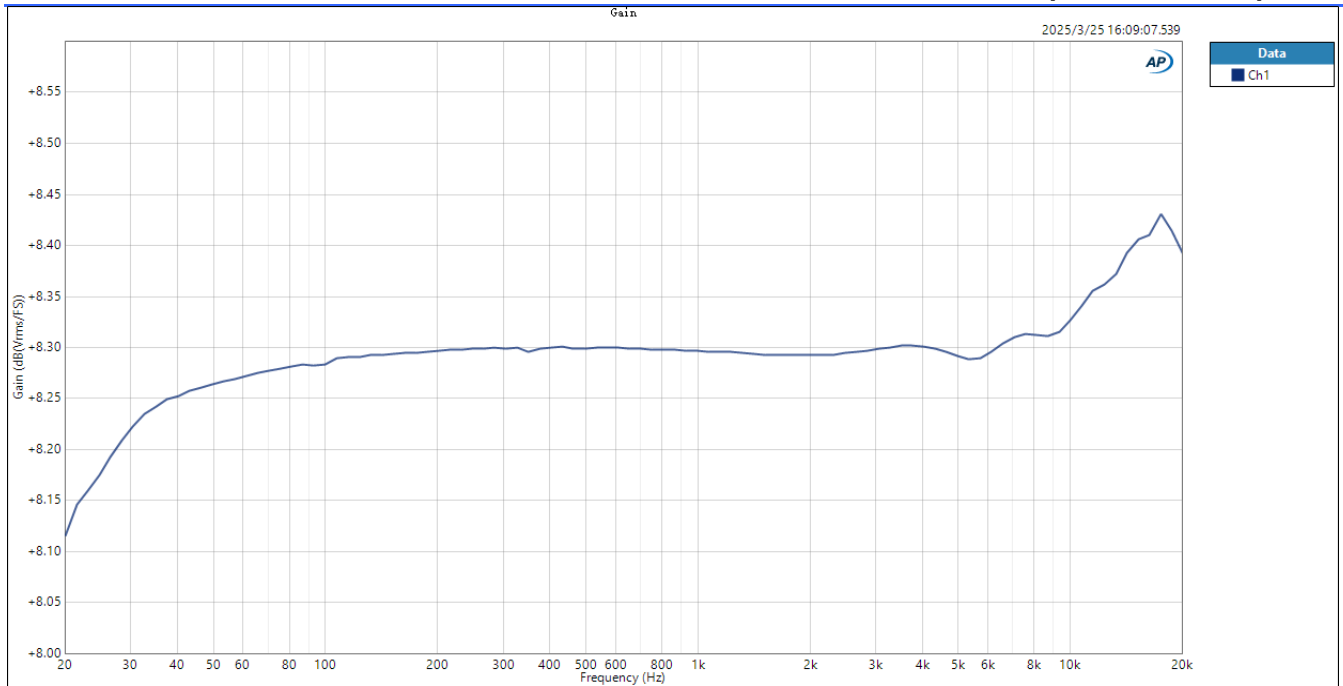
P_O vs η



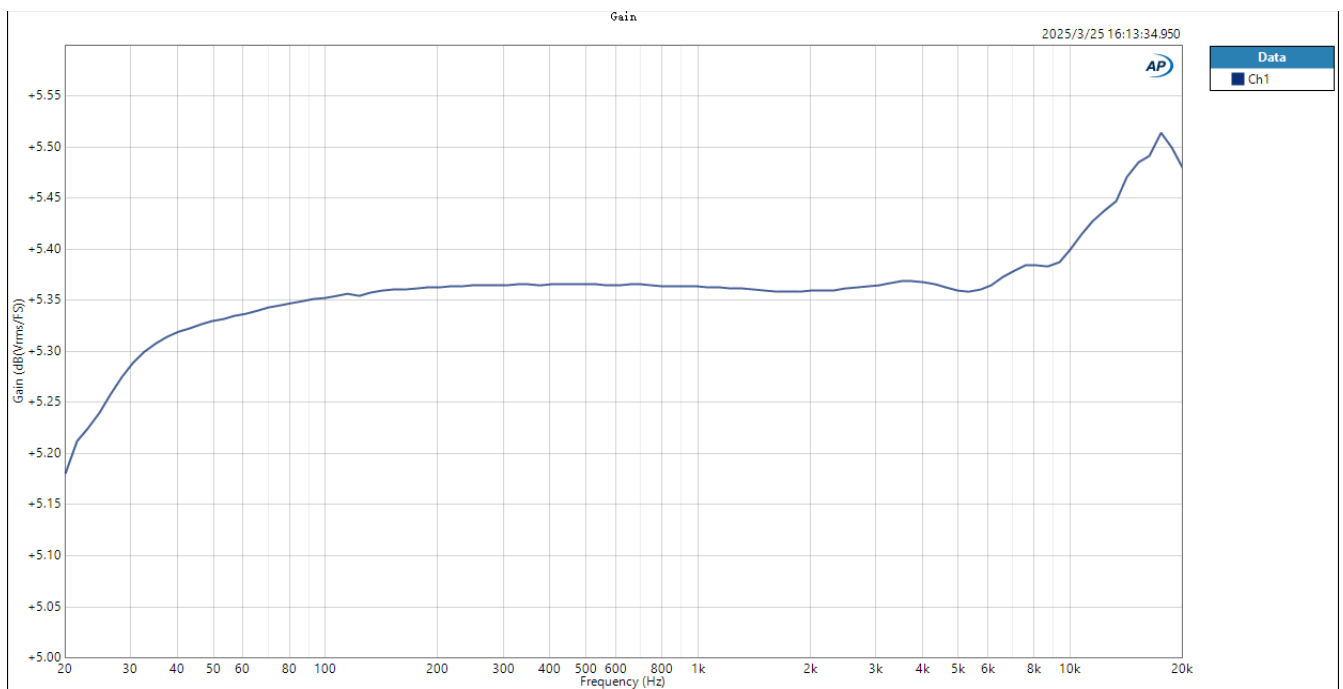
P_O vs η



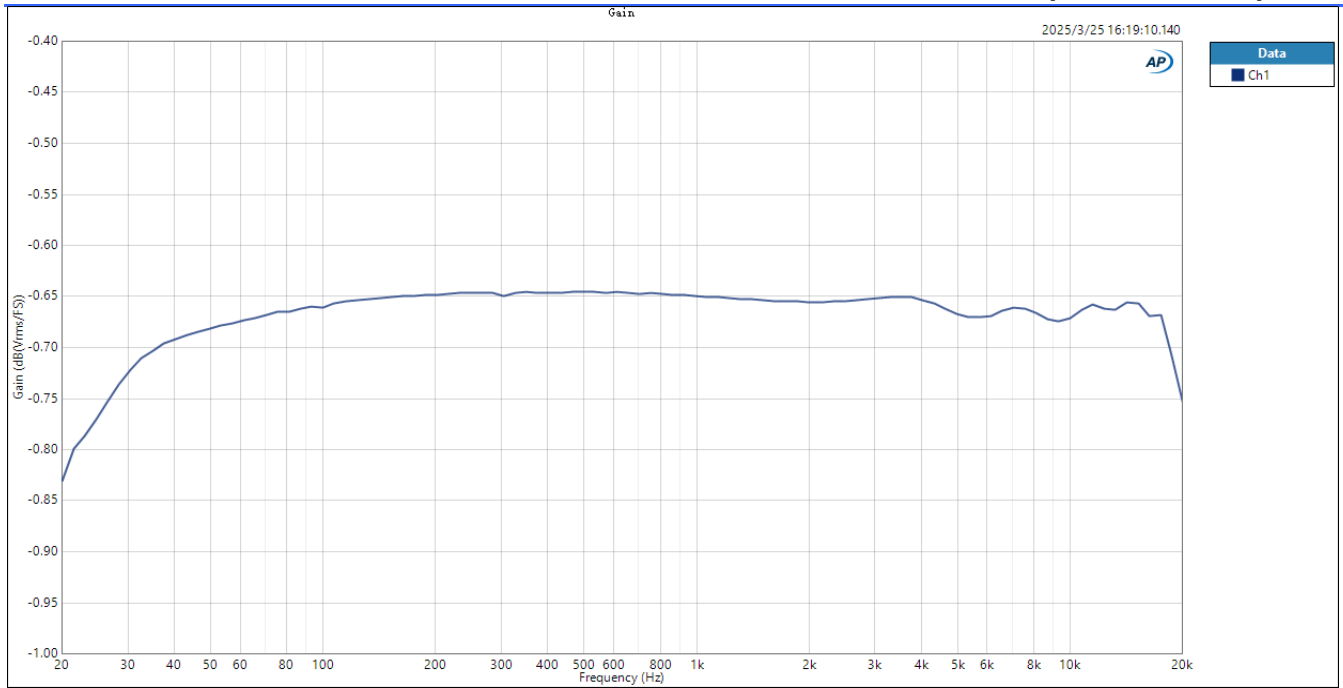




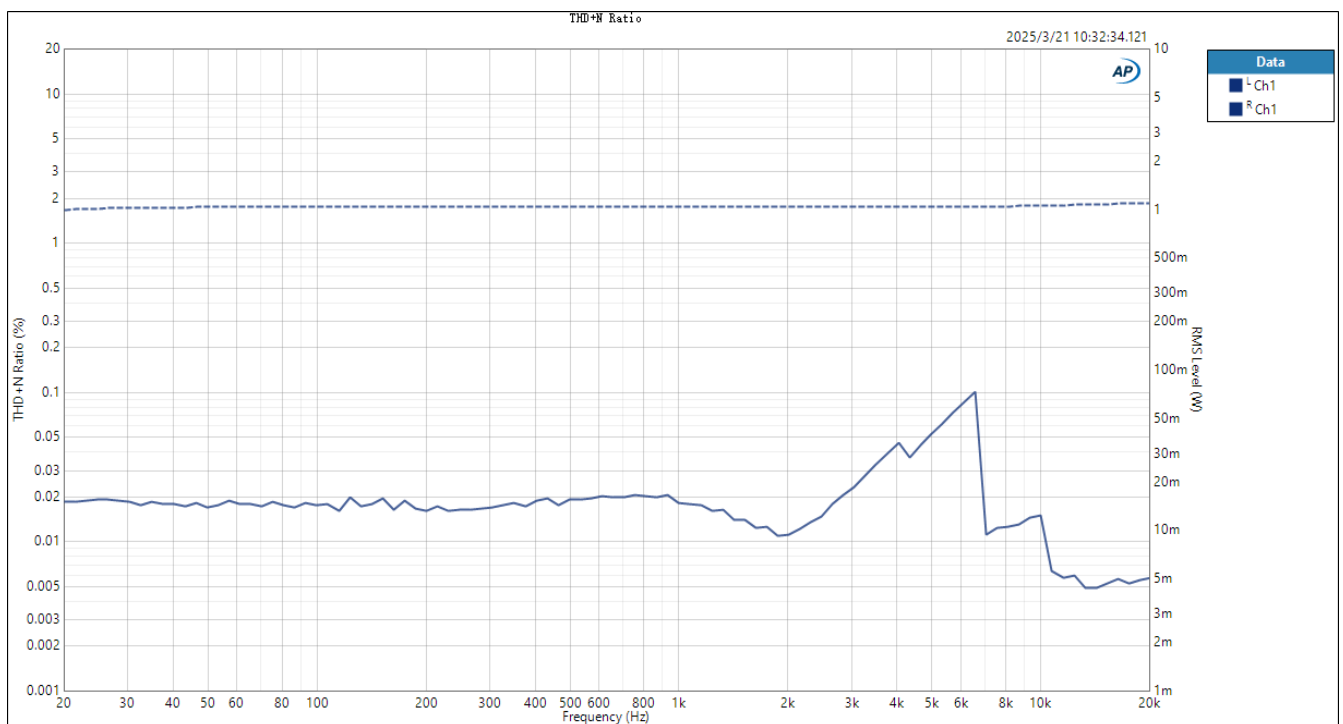
Frequency Respond (VDD = 5V, No Load,,Vin = 0.1Vrms, GAIN/CH = VDD)



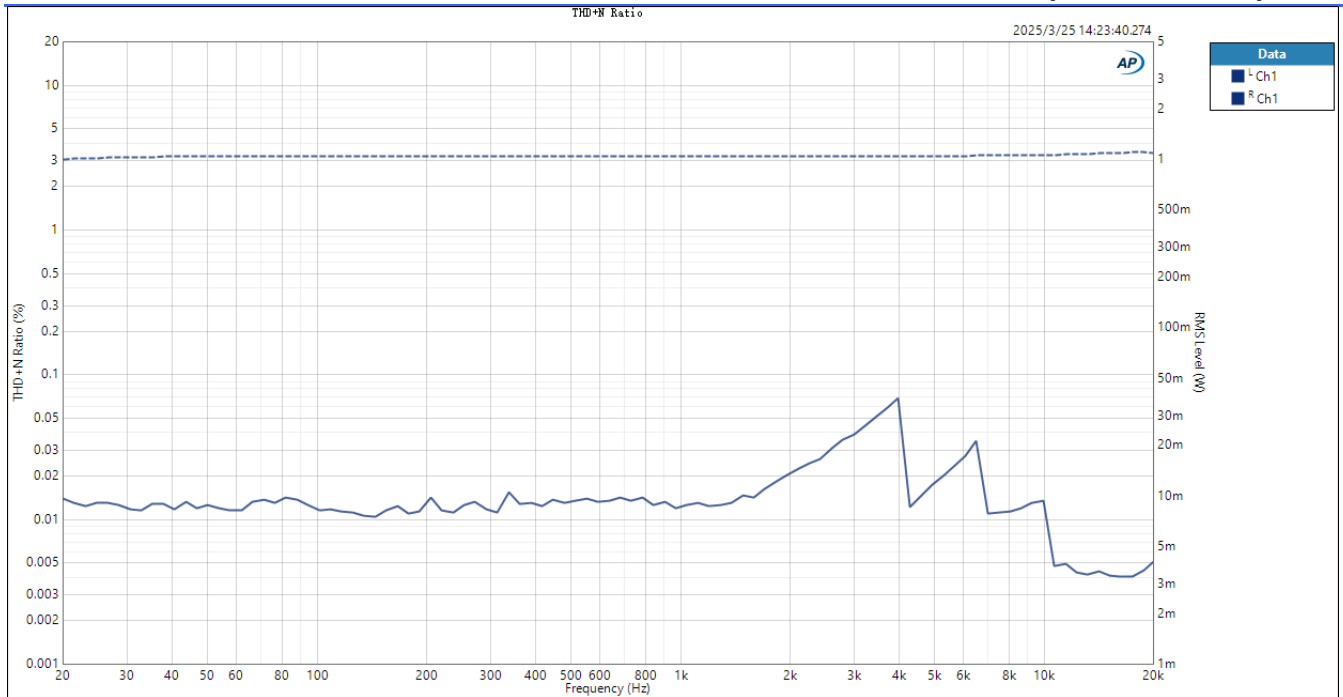
Frequency Respond (VDD = 5V, No Load,,Vin = 0.1Vrms, GAIN/CH = VDD through 100kohm)



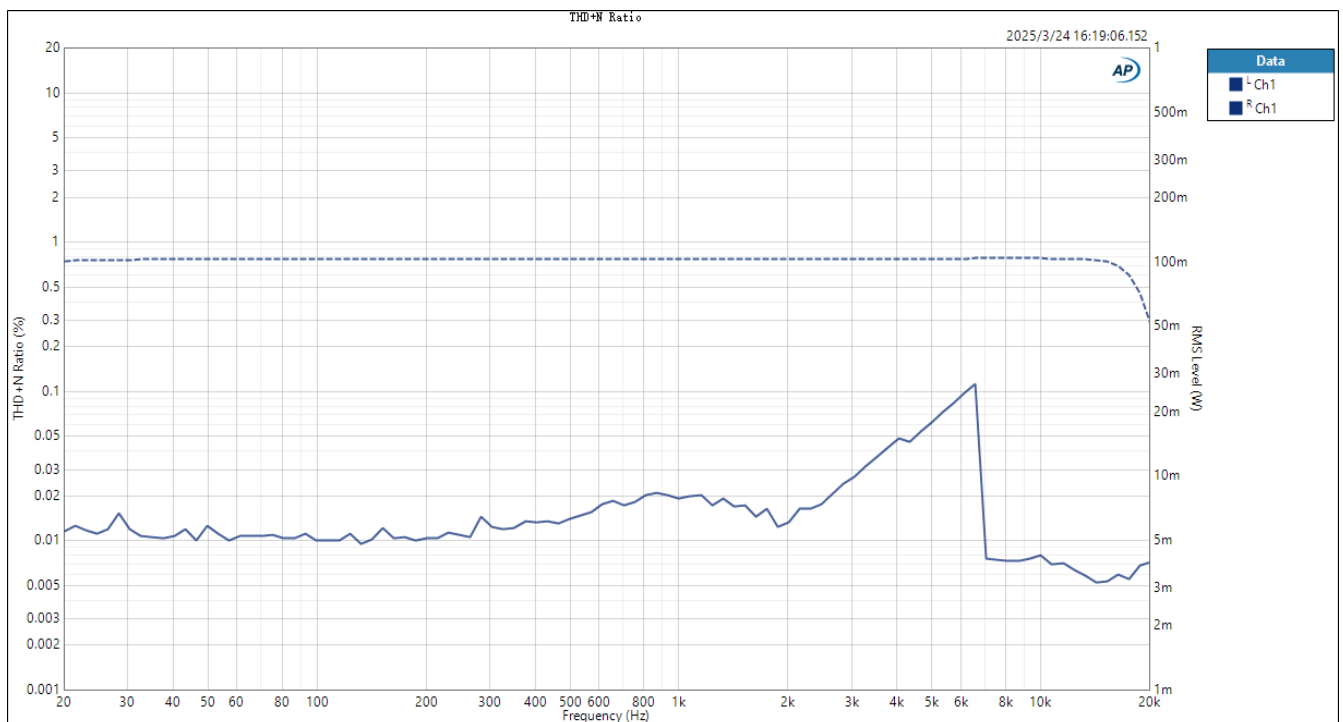
Frequency Respod (VDD = 5V, No Load,,Vin = 0.1Vrms, GAIN/CH = GND through 100kohm)



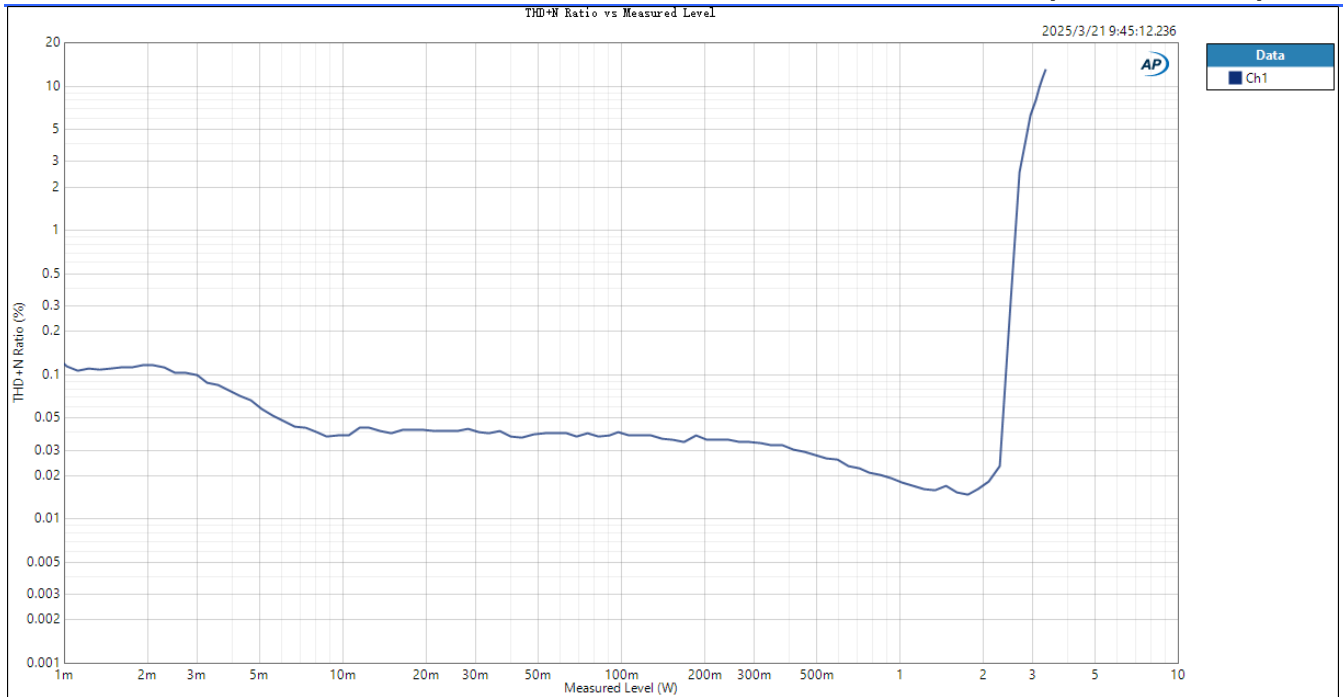
THD+N vs frequency (Po = 1W, Load = 4ohm)



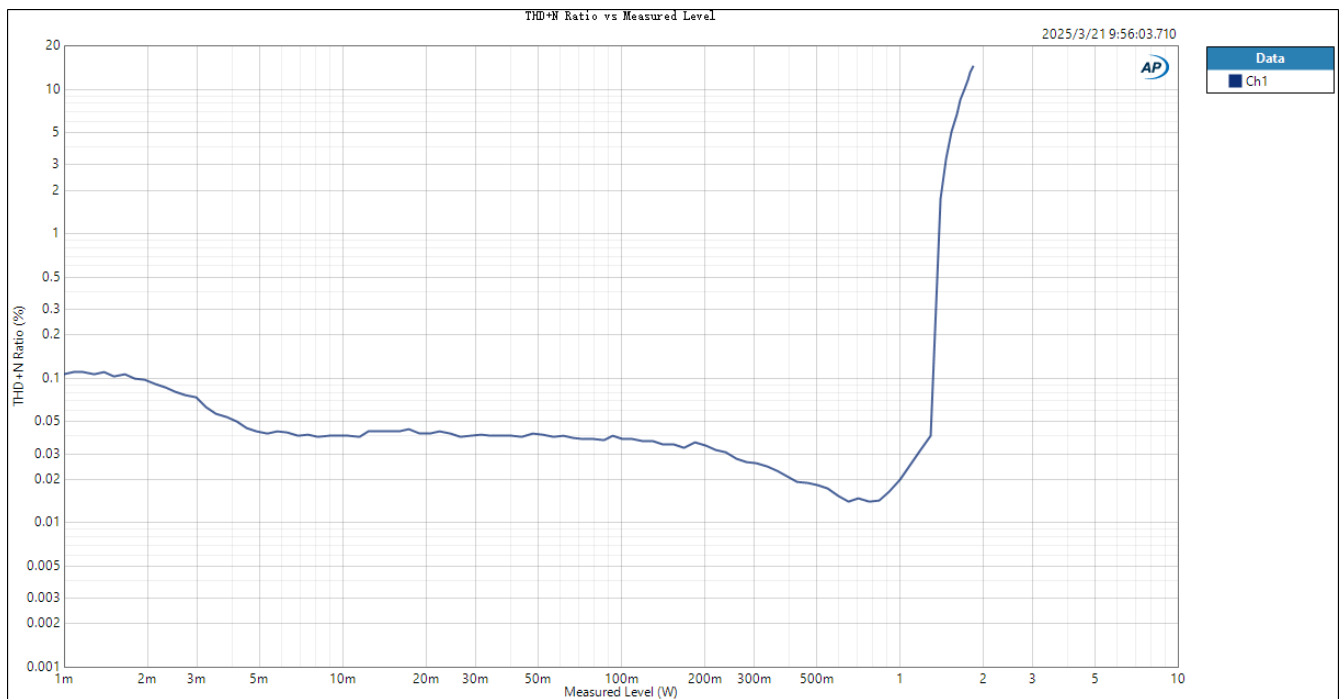
THD+N vs frequency (Po = 1W, Load = 8ohm)



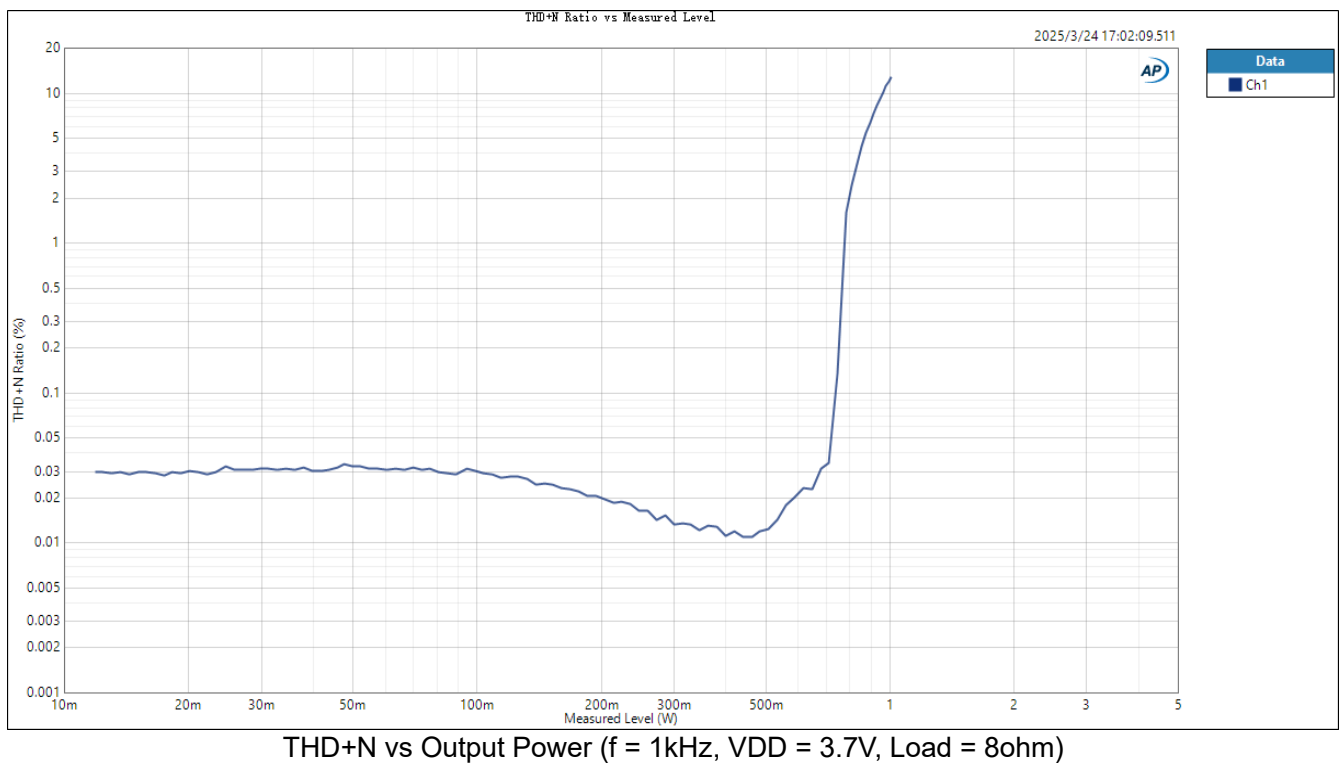
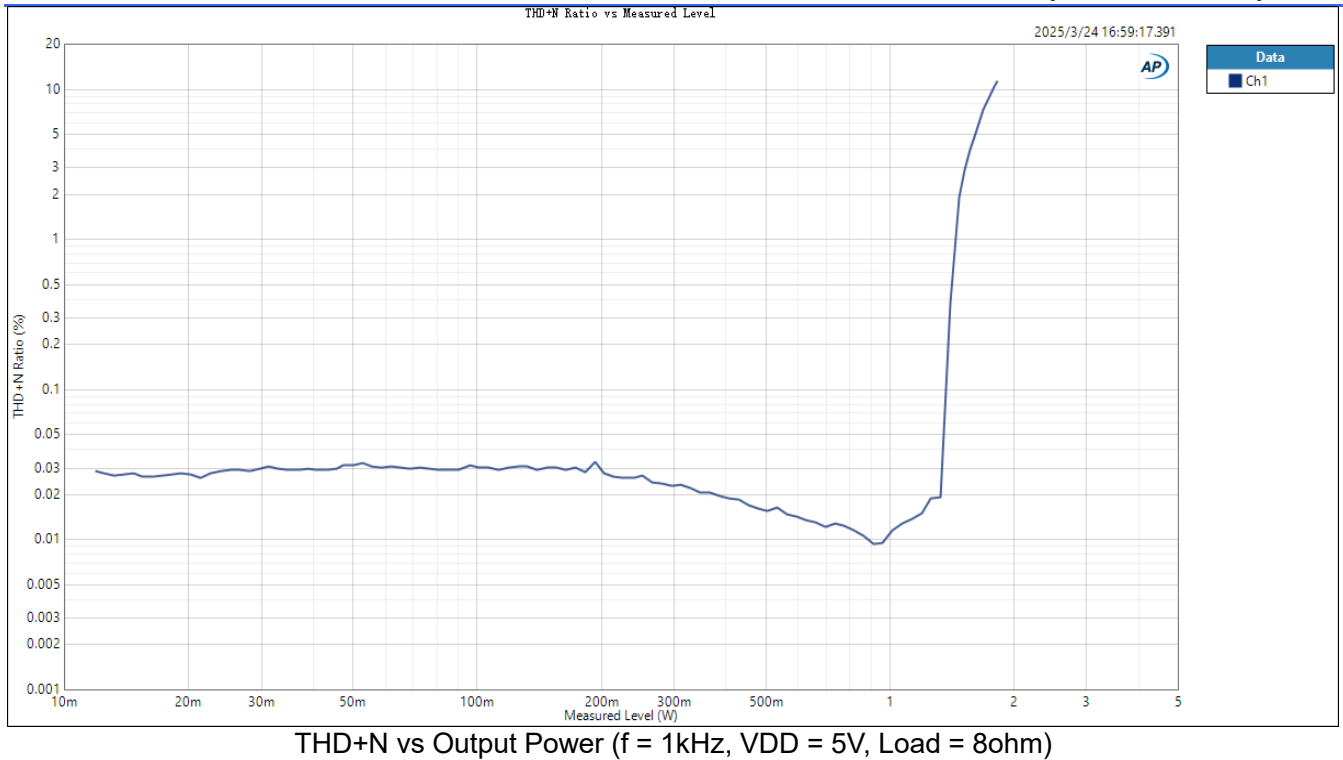
THD+N vs frequency (Po = 100mW, Load = 32ohm)

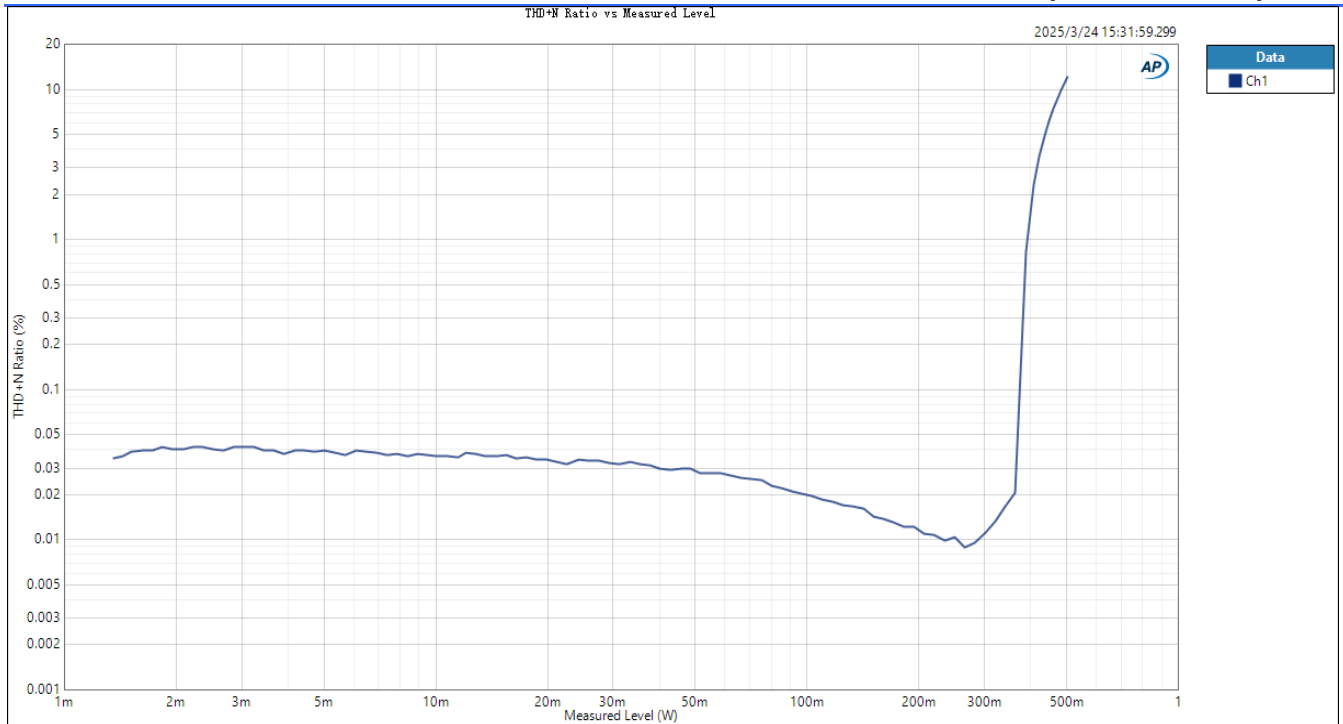


THD+N vs Output Power (f = 1kHz, VDD = 5V, Load = 4ohm)

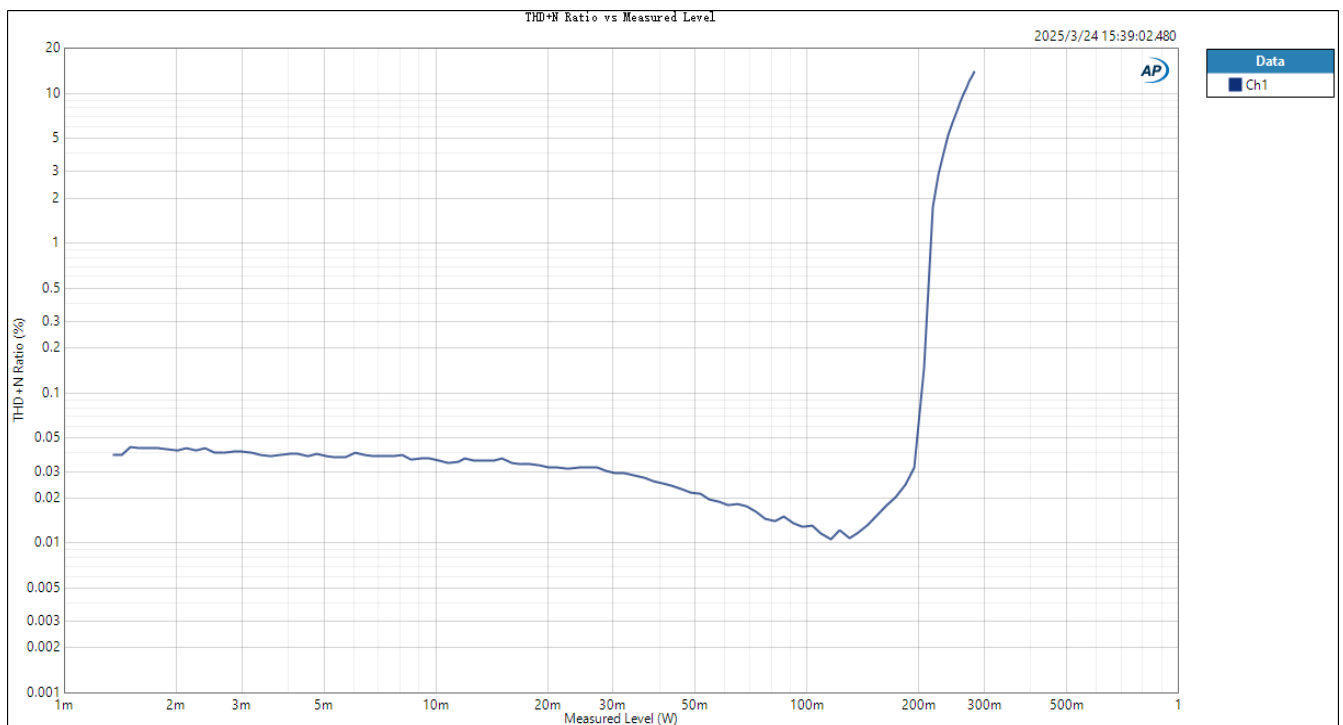


THD+N vs Output Power (f = 1kHz, VDD = 3.7V, Load = 4ohm)





THD+N vs Output Power (f = 1kHz, VDD = 5V, Load = 32ohm)



THD+N vs Output Power (f = 1kHz, VDD = 3.7V, Load = 32ohm)

APPLICATION INFORMATION

The HT517 is a flexible and easy-to-use stereo class-D speaker amplifier with an digital input serial audio port. The HT517 supports a variety of audio clock between 8kHz to 96kHz sample rate.

1 Power Supplies

Only one power supply VDD is required for the HT517. To enable use in a variety of applications, VDD can be operated over a large range of voltages, which is 2.5-5.5V. Bypass VDD with a 0.1uF and 10uF capacitor to GND is required. For 4ohm load and high power application, a larger filtering capacitor is also required.

2 Shut Down Mode (EN pin)

The EN pin is provided to place the device into shutdown. Driving this pin LOW will place the device into shutdown, while pulling it HIGH will bring the device into operation. The shutdown mode is the lowest power consumption mode that all internal blocks are turned off, and the output stage is in a Hi-Z state.

The volume is ramped down when entering shutdown mode.

3 Standby Mode

When the EN pin is high and the DIX pins are not toggling, the device automatically works in standby mode, in which the Class D amplifier is off and the output stage is in a Hi-Z state.

The volume is not ramped down when entering standby mode.

4 Gain Selection (GAIN/CH)

The GAIN/CH pin can select the system gain while in I²S modes, see as the following table.

HT517 是一颗简单易用且灵活的数字输入 D 类音频功放, 其支持 8k~96kHz 的采样频率。

HT517 仅需要一个电源供电, 即 VDD 加 2.5-5.5V。在 VDD 和 GND 间需加一个 0.1uF 和 10uF 滤波电容。对于 4ohm 负载和高功率应用, 更大的滤波电容是需要的。

EN 脚拉低时, 芯片进入关断模式; EN 拉高时, 芯片进入工作状态。在关断模式下, 芯片进入低功耗状态, 内部所有模块关闭, 输出进入高阻状态。

进入关断模式前, 音量是逐渐降低淡出的。

当EN为高, DIX引脚均没有翻转, 器件自动进入待机模式, 此时功放关闭, 输出进入高阻状态。

进入待机模式时, 音量不能逐渐降低淡出。

在 I²S 模式下, 可通过 GAIN/CH 脚, 设置系统增益, 如下表。

Table1 System Gain Selection

GAIN/CH pin	System Gain
Directly connected to GND	14.3dBV
Floating	11.3dBV
Directly connected to VDD	8.3dBV
Connected to VDD thru 100k resistor	5.3dBV
Connected to GND thru 100k resistor	-0.7dBV

5 Serial Audio Port

5.1 Digital Audio Interface (DIX) Mode Selection

The DIX mode selection can be seen as the following table. Take DIX mode A as an example, BCLK is put by DI0, LRCK is put by DI1, SDIN is put by DI2, the left channel of the audio is selected to send to the amplifier output.

Once all the clocks are identified and valid for 4 LRCK periods, the DIX mode selection is latched, and the amplifier is about to turn on.

Once the DIX mode is latched, it won't change until the device is resumed from shutdown mode, undervoltage protection, or standby mode. OTP and OCP do not change the DIX mode.

DIX 模式选择如下表。以模式 A 为例，DI0 接 BCLK，DI1 接 LRCK，SDIN 接 DI2，数字音频中的左通道被放大至功放输出。

当上述所有信号被识别并在 4 个 LRCK 周期内持续有效，DIX 模式被锁定，功放准备开启。

当 DIX 模式被锁定后，除非芯片从关断模式、欠压保护、待机模式切换恢复，否则 DIX 模式不会改变。OTP 和 OCP 不会改变 DIX 模式。

Table2 DIX Mode Selection

BCLK	LRCK	SDIN	DIX Mode	Channel Selection
DI0	DI1	DI2	A	Left
DI1	DI2	DI0	B	Right
DI2	DI0	DI1	C	Left/2+Right/2

5.2 Valid Digital Audio Signal

The GAIN/CH pin can select the system gain while in I²S modes see as the following table.

在 I²S 模式下，可通过 GAIN/CH 脚，设置系统增益，如下表。

Table3 Resolution and Frame Widths

Resolution (Bits)	BCLK periods per LRCK in I2S Mode	BCLK periods per LRCK in TDM mode
16	32	128
24	48	NA
32	64	256

Table4 BCLK Frequencies (kHz)

	I2S mode			TDM Mode	
LRCK	32 BCLK / LRCK	48 BCLK / LRCK	64 BCLK / LRCK	128 BCLK / LRCK	256 BCLK / LRCK
8	256	384	512	1024	2048
16	512	768	1024	2048	4096
32	1024	1536	2048	4096	8192
44.1	1411.2	2116.8	2822.4	5644.8	11289.6
48	1536	2304	3072	6144	12288
88.2	2822.4	4233.6	5644.8	11289.6	22579.2
96	3072	4608	6144	12288	24576

5.3 TDM Mode

Once a frame sync pulse is used for LRCK (LRCK is high for 4 BCLK or less), TDM mode is selected, the device only accepts 8 channel of 16-bit or 32-bit data.

The TDM Channel can be selected as the following table.

一个 LRCK 同步信号 (高 ≤ 4 个 BCLK) 可触发芯片进入 TDM 模式, 芯片只能接受 8 通道 16bit 或 24bit 的 TDM 信号。

TDM 通道可通过下表所示选择。

Table5 TDM Mode Selection

Dlx Mode	GAIN/CH	Selected Channel
A	GND	0
A	VDD	1
A	Floating	2
B	VDD	3
B	GND	4
C	GND	5
C	Floating	6
C	VDD	7

6 DC Blocking Filter

DC content in the audio signal can damage loudspeakers and even small amounts of DC offset in the signal path cause audible artifacts when muting and unmuting the speaker amplifier. For these reasons, the amplifier employs DC blocking methods which is a high-pass filter provided at the front of the data path to remove any DC from incoming audio data before it is presented to the audio path.

The -3 dB corner frequencies for the filter are specified as the following table.

音频信号的直流成分, 可能损坏喇叭, 或者产生输出直流偏置进而在静音/解除静音时产生噪声。因此, HT517 在数据通道前端设置高通滤波器, 以在数据输入端去除直流成分。

该滤波器的截止频率参数如下表。

Table6 TDM Mode Selection

fs	-3dB cutoff frequency of the DC blocking filter
96kHz	3.75Hz
48kHz	3.75Hz
44.1kHz	3.47Hz
8kHz	0.65Hz

7 Closed-Loop Filter-less Class-D Amplifier

The closed-loop filter-less Class-D amplifier offers a much higher efficiency with same level performance than Class-AB amplifier. The switching rate of the amplifier is internally fixed around 300 kHz.

Advanced EMI suppression with spread spectrum control enables fully filter-less or the use of inexpensive ferrite bead filters while meeting EMC requirements for system cost reduction.

7.1 Volume Ramping

The HT517 ramps down the volume before the device goes into shutdown mode, or the power supply is removed.

The volume ramping from original gain to mute takes around 13ms, and it only occurs if BCLK and LRCK remain valid and VDD remains in its operating range for at least 13ms after EN goes low.

7.2 Super Low Pop Noise

The HT517 integrates a comprehensive function to lower the turn-on and turn off pop noise.

8 Speaker Amplifier Protection Suite

The speaker amplifier in the HT517 includes a robust suite of error handling and protection features. It is protected against Over-Current, Under-Voltage, Over-Temperature, and Clock Errors.

8.1 Over-Current Protection

The HT517 features over-current conditions against the output stage short-circuit conditions. The amplifier outputs are switched to a Hi-Z state when the short circuit protection latch is triggered. The device will automatically attempt to resume after around 100us. If the over-current condition is still not cleared, the device will again go into protection.

8.2 Over-temperature Protection

Over-temperature protection on the HT517 device prevents damage to the device when the internal die temperature exceeds 150°C. This triggering point has a $\pm 15^\circ\text{C}$ tolerance from device to device. Once the die temperature exceeds the thermal triggering point, the device is switched to the shutdown state and the outputs are disabled. The device will automatically attempt to resume after around 100us. If the over-temperature condition is still not cleared, the device will again go into protection.

8.3 Under-voltage Protection

When the voltage on PVDD pin falls below the under-voltage threshold, the UVP circuit puts the device into shutdown mode. The device recovers automatically once the under-voltage condition has been removed.

芯片内置的闭环免滤波 D 类功放具有相比 AB 类效率更高且音频性能相似的性能。D 类功放的开关频率被固定在 480kHz 附近。

芯片具有先进的扩频功能来一直 EMI，使其可以完全免滤波或使用价格低廉且小体积铁氧体磁珠即可满足 EMC 要求。

HT517 具有淡出功能，在其关断、无电源前，音量会逐渐降低至静音。

该淡出功能从原始音量倒静音需要约 13ms，EN 拉低后 BCLK 和 LRCK 保持有效、VDD 在工作范围内持续 13ms，带出功能才能完全有效。

HT517 集成了全面的抑制开关机 pop 声功能，具有极其优越的 pop 声。

HT517 具有多种保护功能，包括过流、欠压、过温、时钟错误等保护。

HT517 输出级短路时，发生了过流，此时芯片进入保护状态，功放输出切换到高阻状态。经过约 100us 时间后，芯片将自动尝试恢复，若过流状态已消失，芯片恢复；若过流状态仍在，芯片再次进入保护状态。

过温保护在芯片内部结温达到 150°C 时发生，以防止芯片损坏，此时芯片进入关断状态。经过约 100us 时间后，芯片将自动尝试恢复，若过温状态已消失，芯片恢复；若过温状态仍在，芯片再次进入保护状态。

当 PVDD 电压低于欠压保护点 ($U_{VE_{FTH}}$) 时，芯片进入保护状态，芯片关闭。当 PVDD 高于欠压保护点 ($U_{VE_{RTH}}$) 后，芯片立即自动恢复。

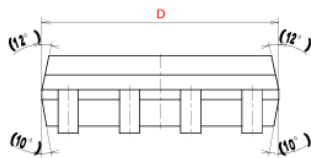
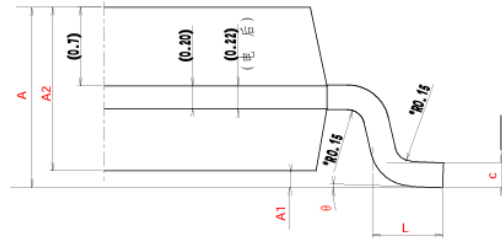
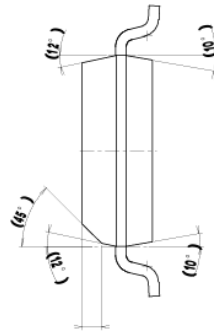
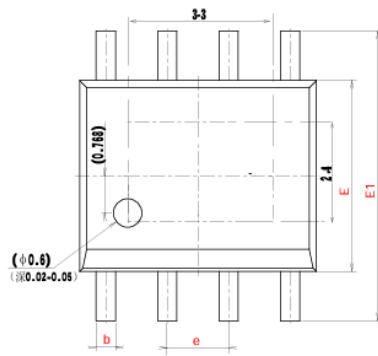
8.4 Clock error detection

When any clock of BCLK, LRCK halt or invalid, the audio amplifier is turned off immediately without volume ramping, the output stage goes into a Hi-Z state. If there's still toggling on D1x pins, the D1x mode selection is re-detected; if not, the device goes into standby mode.

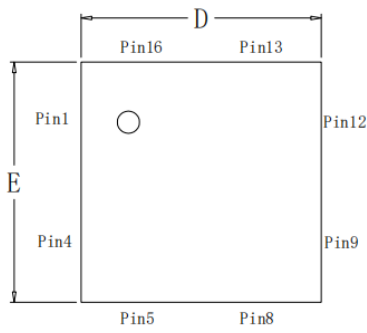
当 SCLK、LRCK 停止或无效时，功放立刻关闭，无淡出，输出高阻。如果此时 D1x 仍有信号，D1x 模式将被重新检测识别；如果没有，芯片进入待机模式。

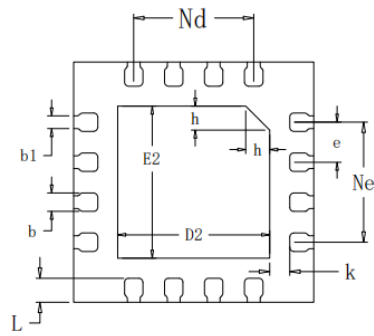
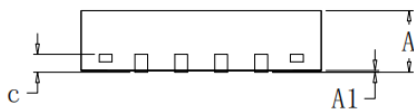
■ PACKAGE OUTLINE

ESOP8

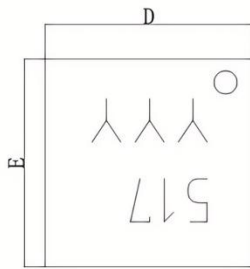
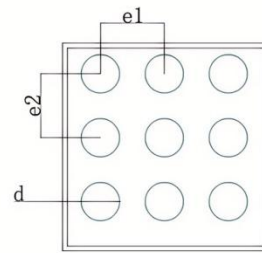
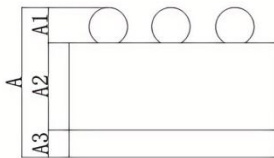


字符	Dimension millimeters		
	Min	Standard	Max
A	1.350	1.500	1.700
A1	0	0.050	0.150
A2	1.350	1.450	1.550
b	0.360	0.400	0.440
c	0.215	0.220	0.235
D	4.800	4.900	5.000
E	3.840	3.940	4.040
E1	5.900	6.000	6.100
e	1.27BSC		
L	0.400	0.550	0.700
⊖	0°		8°

QFN3×3-16L

Top View

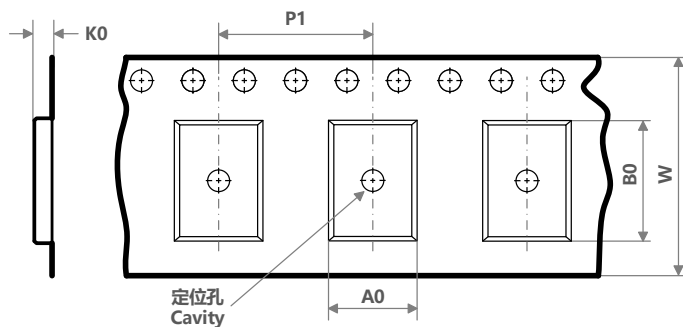
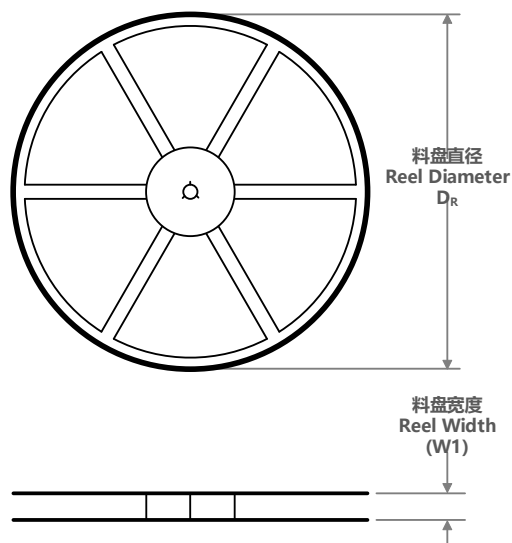
Side View

Bottom View

Side View

SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	0.70	0.75	0.80
A1	--	0.02	0.05
b	0.18	0.23	0.28
b1	0.11	0.16	0.21
c	0.203 REF		
D	2.90	3.00	3.10
D2	1.80	1.90	2.00
Nd	1.50 BSC		
e	0.50 BSC		
E	2.90	3.00	3.10
E2	1.80	1.90	2.00
Ne	1.50 BSC		
h	0.25	0.30	0.35
k	0.20	0.25	0.30
L	0.25	0.30	0.35

WLCSP9
Top View

Bottom View

Side View


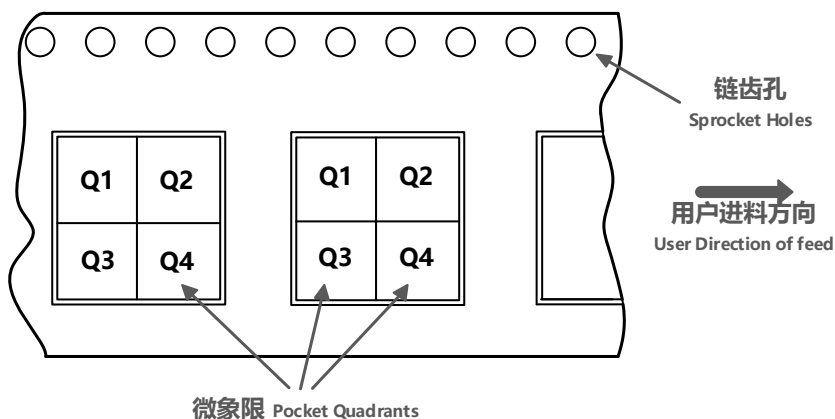
Symbol	NOM	Toler.
A	0.614	±0.038
A1	0.194	±0.020
A2	0.380	±0.013
A3	0.040	±0.005
D	1.275	±0.025
E	1.275	±0.025
d	0.268	±0.020
e1	0.400	NA
e2	0.400	NA

■ TAPE AND REEL INFORMATION

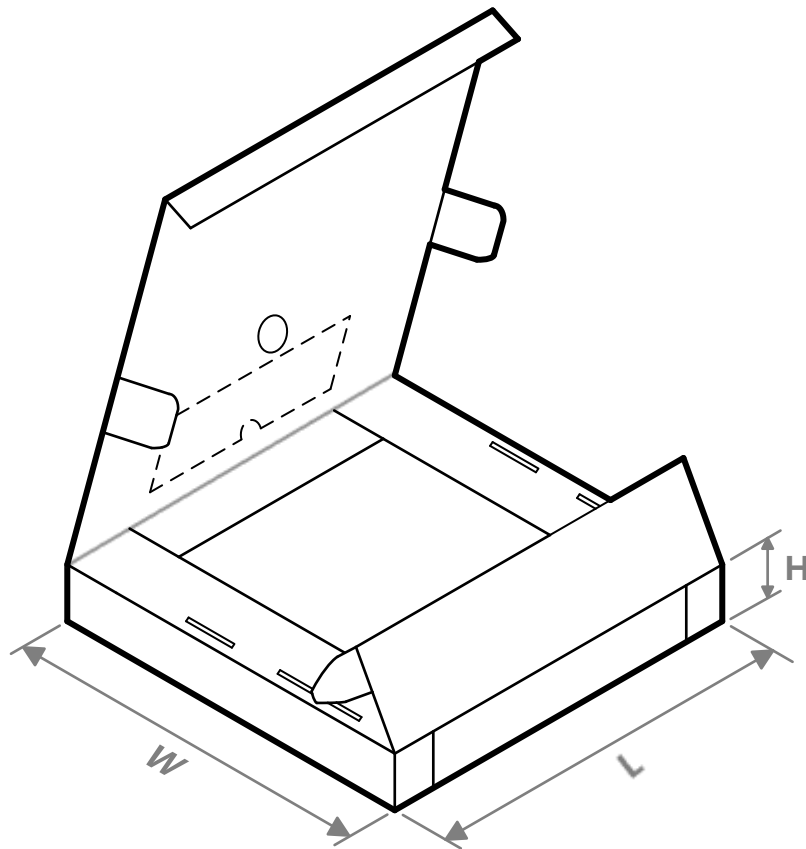


A0	Dimension designed to accommodate the component width; 料槽宽度
B0	Dimension designed to accommodate the component length; 料槽长度
K0	Dimension designed to accommodate the component thickness; 料槽厚度
W	Overall width of the carrier tape; 载带整体宽度
P1	Pitch between successive cavity centers; 相邻槽中心间距

编带 PIN1 方位象限分配 Quadrant Assignments for Pin1 Orientation in Tape



器件料号 Part No.	封装类型 Package Type	封装标识 Package Abbr.	引脚数 Pins	SPQ	料盘直径 D_R (mm)	料盘宽度 W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 象限 Quadrant
HT517WCSR	WLCSP9	WCS	9	5000	330	9.5	1.38	1.38	0.74	4	8	Q1
HT517SQER	QFN3×3	SQE	16	5000	330	12	3.3	3.3	1.05	8	12	Q1
HT517SPER	ESOP	SPE	8	2500	330	12	6.55	5.55	1.95	8	12	Q1

TAPE AND REEL BOX INFORMATION


器件料号 Part No.	封装类型 Package Type	封装标识 Package Abbr.	引脚数 Pins	SPQ	长度 Length (mm)	宽度 Width (mm)	高度 Height (mm)
HT517WCSR	WLCSP	WCS	8	5000	345	365	55
HT517SQER	QFN3×3	SQE	16	10000	360	345	65
HT517SPER	ESOP	SPE	8	5000	360	345	65

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